AD 502 – Machine Learning

Pre-Reading Material – Lecture 33

Basic Ensemble Techniques: Voting

Introduction

Max Voting, also known as Majority Voting, is a basic ensemble learning technique used mainly for classification tasks. It combines the predictions of multiple classifiers and selects the class receiving the highest number of votes.

The underlying idea is that a collective decision from several classifiers may be more reliable than the prediction of an individual classifier.

Working Principle

Max Voting follows these steps:

1. Train multiple classification models on the training dataset.
2. Provide the same test observation to every trained model.
3. Collect the predicted class label from each model.
4. Count the votes received by each class.
5. Select the class with the maximum number of votes as the final prediction.

Example of Max Voting

Suppose three classifiers are used to predict whether an email is spam.

Classifier	Predicted class
Decision Tree	Spam
Logistic Regression	Not Spam
Support Vector Machine	Spam

The vote count is:

- Spam: 2 votes
- Not Spam: 1 vote

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Therefore, the final prediction is Spam, because it receives the majority of votes.

Hard Voting and Soft Voting

There are two commonly discussed voting approaches.

Hard voting: Each classifier contributes one vote for its predicted class. The class receiving the most votes is selected.

Soft voting: Each classifier provides class probabilities. These probabilities are combined, usually by averaging them, and the class with the highest combined probability is selected.

For example, if a model predicts a 70% probability of spam and another predicts 80%, soft voting considers these confidence values rather than only the predicted labels.

Soft voting is particularly useful when the classifiers produce meaningful probability estimates.

Advantages of Max Voting

- Easy to understand and implement.
- Combines predictions from different classifiers.
- Can reduce the influence of an individual classifier's incorrect prediction.
- Can improve classification performance when the models complement one another.

Limitations of Max Voting

- It may not improve performance when all classifiers make similar errors.
- In hard voting, every classifier receives equal voting weight.
- A tie may occur when classes receive equal numbers of votes.
- Hard voting does not directly consider prediction confidence.

7. Applications

Max Voting can be used for spam detection, disease classification, sentiment analysis, fraud detection, and image classification.

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For example, an ensemble of three classifiers can combine their decisions to classify a customer review as positive or negative.

Summary

Max Voting combines the class predictions of multiple classifiers and selects the class with the most votes. Hard voting uses predicted class labels, whereas soft voting combines class probabilities. It is a simple and effective ensemble technique when the individual classifiers provide useful and complementary predictions.

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